

EXP 1:

Code:

```
experiment1.py - C:/Users/ayesh/Downloads/query lab/experiment1.py (3.13.14)
File Edit Format Run Options Window Help

import pandas as pd

# Creating the DataFrame
data = {
    "DEPARTMENT_ID": [10, 20, 30, 40, 50, 60, 70, 80, 90, 100,
                     110, 120, 130, 140, 150, 160, 170, 180,
                     190, 200, 210, 220, 230, 240, 250, 260, 270],
    "DEPARTMENT_NAME": [
        "Administration", "Marketing", "Purchasing", "Human Resources",
        "Shipping", "IT", "Public Relations", "Sales", "Executive",
        "Finance", "Accounting", "Treasury", "Corporate Tax",
        "Control And Credit", "Shareholder Services", "Benefits",
        "Manufacturing", "Construction", "Contracting", "Operations",
        "IT Support", "NOC", "IT Helpdesk", "Government Sales",
        "Retail Sales", "Recruiting", "Payroll"
    ],
    "MANAGER_ID": [
        200, 201, 114, 203, 121, 103, 204, 145, 100,
        108, 205, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0
    ],
    "LOCATION_ID": [
        1700, 1800, 1700, 2400, 1500, 1400, 2700, 2500, 1700,
        1700, 1700, 1700, 1700, 1700, 1700, 1700, 1700, 1700,
        1700, 1700, 1700, 1700, 1700, 1700, 1700, 1700
    ]
}

# Create DataFrame
df = pd.DataFrame(data)

# Display the DataFrame
print("Employee Department Data:")
print(df)

# Select distinct Department IDs
distinct_ids = df["DEPARTMENT_ID"].drop_duplicates()

# Display the distinct Department IDs
print("\nDistinct Department IDs:")
print(distinct_ids)
```

Output:

```
*IDLE Shell 3.13.14*
File Edit Shell Debug Options Window Help

Python 3.13.14 (tags/v3.13.14:fd17997, Jun 10 2026, 13:03:48) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

>>>
===== RESTART: C:/Users/ayesh/Downloads/query lab/experiment1.py =====
Employee Department Data:
  DEPARTMENT_ID DEPARTMENT_NAME  MANAGER_ID  LOCATION_ID
0             10  Administration         200           1700
1             20     Marketing         201           1800
2             30     Purchasing         114           1700
3             40  Human Resources         203           2400
4             50     Shipping         121           1500
5             60              IT         103           1400
6             70  Public Relations         204           2700
7             80         Sales         145           2500
8             90     Executive         100           1700
9            100         Finance         108           1700
10           110     Accounting         205           1700
11           120         Treasury          0           1700
12           130     Corporate Tax          0           1700
13           140  Control And Credit          0           1700
14           150  Shareholder Services          0           1700
15           160         Benefits          0           1700
16           170     Manufacturing          0           1700
17           180     Construction          0           1700
18           190     Contracting          0           1700
19           200     Operations          0           1700
20           210     IT Support          0           1700
21           220             NOC          0           1700
22           230     IT Helpdesk          0           1700
23           240   Government Sales          0           1700
24           250     Retail Sales          0           1700
25           260     Recruiting          0           1700
26           270         Payroll          0           1700
```

Distinct Department IDs:

0	10
1	20
2	30
3	40
4	50
5	60
6	70
7	80
8	90
9	100
10	110
11	120
12	130
13	140
14	150
15	160
16	170
17	180
18	190
19	200
20	210
21	220
22	230
23	240
24	250
25	260
26	270

Name: DEPARTMENT_ID, dtype: int64

>>>

EXP 2:

Code:

```
experiment2.py - C:/Users/ayesh/Downloads/query lab/experiment2.py (3.11.14)
File Edit Format Run Options Window Help
# Import pandas library
import pandas as pd

# Read the CSV file
df = pd.read_csv("job_history.csv")

# Display all records from the job history file
print("JOB HISTORY DATA")
print(df)

# Print a separator line
print("-" * 50)

# Find employees who have done two or more jobs
employee_ids = df["EMPLOYEE_ID"].value_counts()
multiple_jobs = employee_ids[employee_ids >= 2]

# Display the employee IDs
print("EMPLOYEE IDS WHO HAVE DONE TWO OR MORE JOBS")
print(multiple_jobs)

# Print a separator line
print("-" * 50)

# Display the total number of employees
print("Total Employees with Two or More Jobs:",
      multiple_jobs.count())
```

Output:

```
JOB HISTORY DATA
  EMPLOYEE_ID  START_DATE  END_DATE  JOB_ID  DEPARTMENT_ID
0          102  2001-01-13  2006-07-24   IT_PROG             60
1           101  1997-09-21  2001-10-27  AC_ACCOUNT          110
2           101  2001-10-28  2005-03-15   AC_MGR            110
3           201  2004-02-17  2007-12-19   MK_REP             20
4           114  2006-03-24  2007-12-31  ST_CLERK             50
5           122  2007-01-01  2007-12-31  ST_CLERK             50
6           200  1995-09-17  2001-06-17  AD_ASST             90
7           176  2006-03-24  2006-12-31   SA_REP             80
8           176  2007-01-01  2007-12-31   SA_MAN             80
9           200  2002-07-01  2006-12-31  AC_ACCOUNT          90

-----
EMPLOYEE IDS WHO HAVE DONE TWO OR MORE JOBS
EMPLOYEE_ID
101      2
200      2
176      2
Name: count, dtype: int64
-----
Total Employees with Two or More Jobs: 3
```

EXP 3:

Code:

```
experiment3.py - C:/Users/ayesh/Downloads/query lab/experiment3.py (3.13.14)
File Edit Format Run Options Window Help

# Import pandas library
import pandas as pd

# Read the CSV file
df = pd.read_csv("jobs.csv")

# Display all records from the jobs file
print("JOBS DATA")
print(df)

# Print a separator line
print("-" * 50)

# Sort the jobs in descending order of JOB_TITLE
sorted_jobs = df.sort_values(by="JOB_TITLE", ascending=False)

# Display the sorted data
print("JOBS IN DESCENDING ORDER OF JOB TITLE")
print(sorted_jobs)

# Print a separator line
print("-" * 50)

# Display the total number of jobs
print("Total Number of Jobs:", sorted_jobs.count()["JOB_ID"])
```

Output:

```
IDLE Shell 3.13.14
File Edit Shell Debug Options Window Help

===== RESTART: C:/Users/ayesh/Downloads/query lab/experiment3.py =====
JOBS DATA
  JOB_ID  JOB_TITLE  MIN_SALARY  MAX_SALARY
0      AD_PRES      President      20080      40000
1      AD_VP  Administration Vice President      15000      30000
2      AD_ASST  Administration Assistant      3000      6000
3      FI_MGR      Finance Manager      8200      16000
4      FI_ACCOUNT      Accountant      4200      9000
5      AC_MGR      Accounting Manager      8200      16000
6      AC_ACCOUNT      Public Accountant      4200      9000
7      SA_MAN      Sales Manager      10000      20080
8      SA_REP      Sales Representative      6000      12008
9      PU_MAN      Purchasing Manager      8000      15000
10     PU_CLERK      Purchasing Clerk      2500      5500
11     ST_MAN      Stock Manager      5500      8500
12     ST_CLERK      Stock Clerk      2008      5000
13     SH_CLERK      Shipping Clerk      2500      5500
14     IT_PROG      Programmer      4000      10000
15     MK_MAN      Marketing Manager      9000      15000
16     MK_REP      Marketing Representative      4000      9000
17     HR_REP      Human Resources Representative      4000      9000
18     PR_REP      Public Relations Representative      4500      10500

-----
JOBS IN DESCENDING ORDER OF JOB TITLE
  JOB_ID  JOB_TITLE  MIN_SALARY  MAX_SALARY
11     ST_MAN      Stock Manager      5500      8500
12     ST_CLERK      Stock Clerk      2008      5000
13     SH_CLERK      Shipping Clerk      2500      5500
8      SA_REP      Sales Representative      6000      12008
7      SA_MAN      Sales Manager      10000      20080
9      PU_MAN      Purchasing Manager      8000      15000
10     PU_CLERK      Purchasing Clerk      2500      5500
18     PR_REP      Public Relations Representative      4500      10500
6      AC_ACCOUNT      Public Accountant      4200      9000
14     IT_PROG      Programmer      4000      10000
0      AD_PRES      President      20080      40000
16     MK_REP      Marketing Representative      4000      9000
15     MK_MAN      Marketing Manager      9000      15000
17     HR_REP      Human Resources Representative      4000      9000
3      FI_MGR      Finance Manager      8200      16000
1      AD_VP  Administration Vice President      15000      30000
2      AD_ASST  Administration Assistant      3000      6000
5      AC_MGR      Accounting Manager      8200      16000
4      FI_ACCOUNT      Accountant      4200      9000

-----
Total Number of Jobs: 19
>>>
```

EXP 4:

Code:

```
# Import pandas and matplotlib
import pandas as pd
import matplotlib.pyplot as plt

# Read the CSV file
df = pd.read_csv("alphabet_stock_data.csv")

# Convert Date column to datetime
df["Date"] = pd.to_datetime(df["Date"])

# Select data between two dates
start_date = "2020-01-03"
end_date = "2020-01-14"

filtered_data = df[(df["Date"] >= start_date) &
                   (df["Date"] <= end_date)]

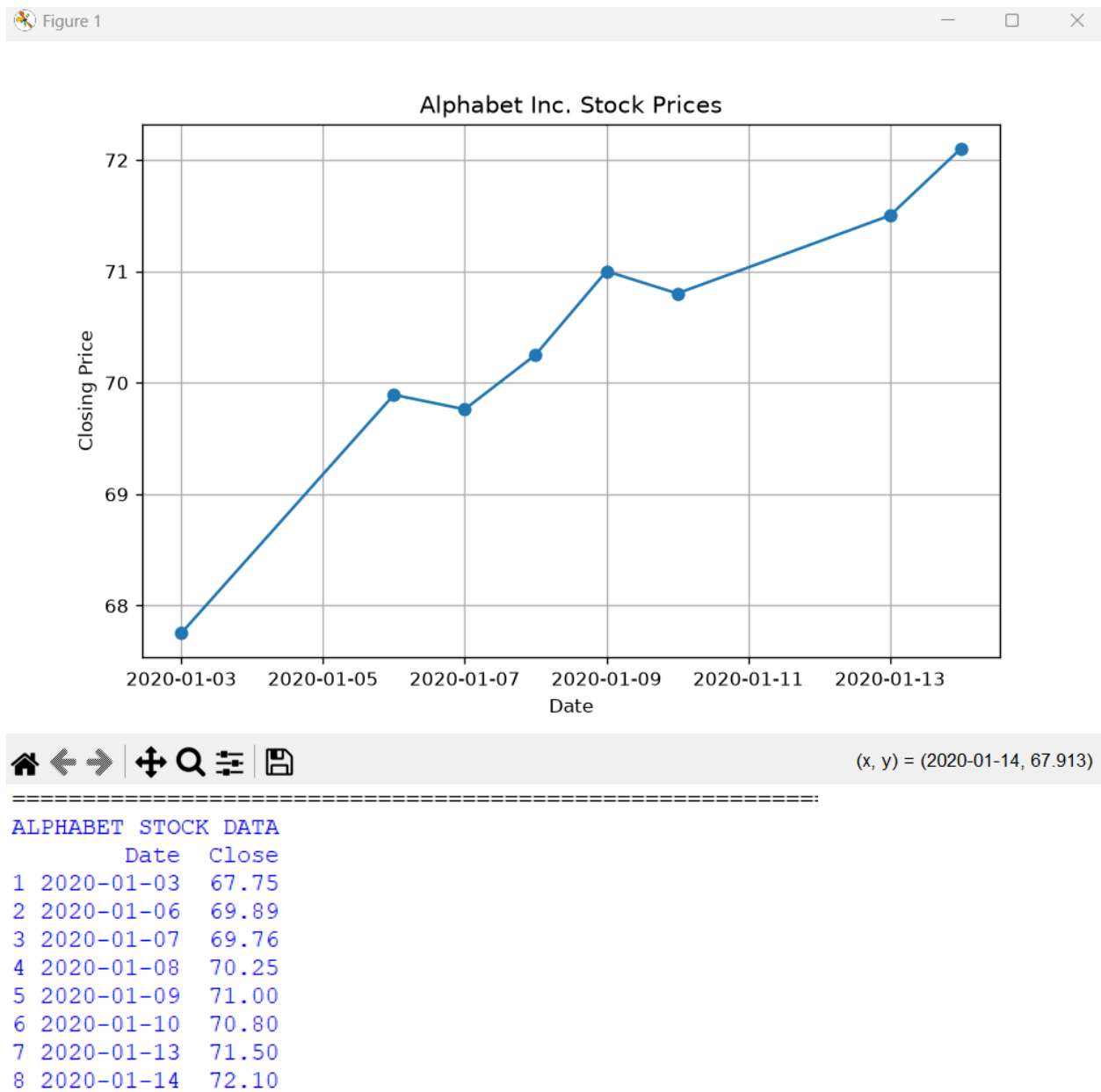
# Display the filtered data
print("ALPHABET STOCK DATA")
print(filtered_data)

# Plot line graph
plt.figure(figsize=(8,5))
plt.plot(filtered_data["Date"],
         filtered_data["Close"],
         marker='o')

plt.title("Alphabet Inc. Stock Prices")
plt.xlabel("Date")
plt.ylabel("Closing Price")
plt.grid(True)

# Display the graph
plt.show()
|
```

Output:



EXP 5:

Code:

```
experiment5.py - C:/Users/ayesh/Downloads/query lab/experiment5.py (3.13.14)
File Edit Format Run Options Window Help
# Import pandas and matplotlib
import pandas as pd
import matplotlib.pyplot as plt

# Read the CSV file
df = pd.read_csv("alphabett_stock_data.csv")

# Convert Date column to datetime
df["Date"] = pd.to_datetime(df["Date"])

# Select data between two dates
start_date = "2020-01-03"
end_date = "2020-01-14"

filtered_data = df[(df["Date"] >= start_date) &
                  (df["Date"] <= end_date)]

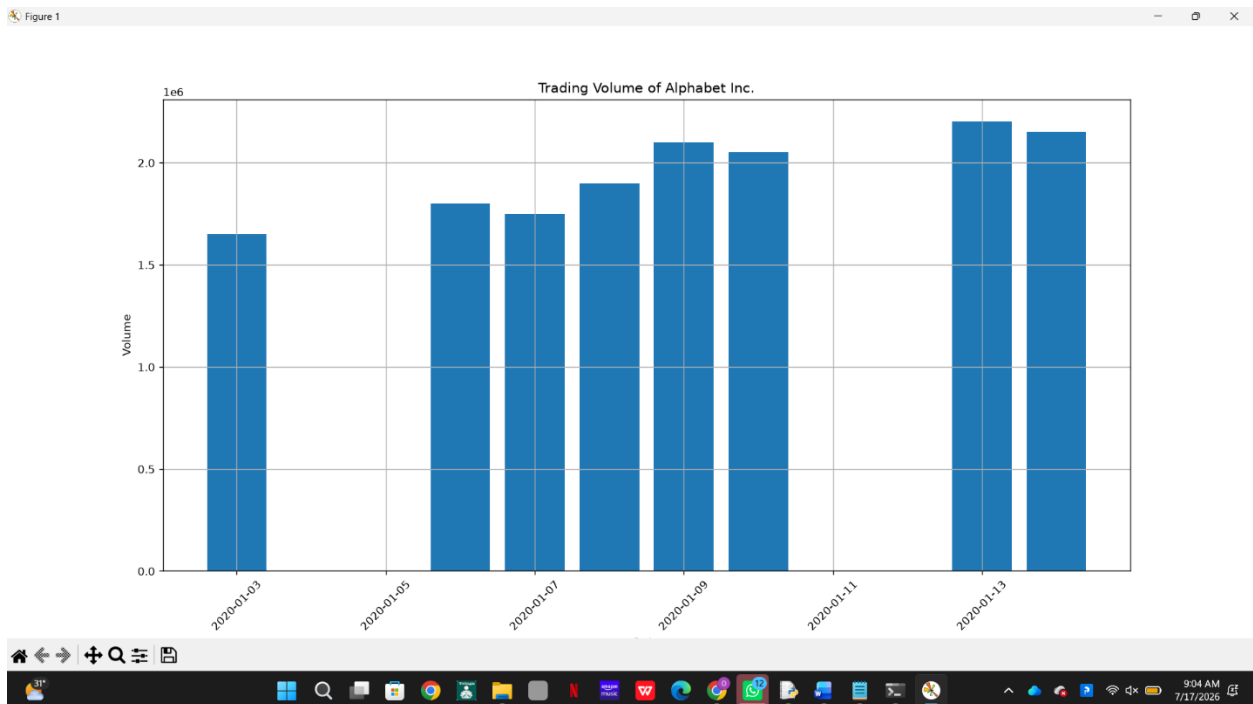
# Display the filtered data
print("ALPHABET STOCK DATA")
print(filtered_data)

# Create bar plot
plt.figure(figsize=(8,5))
plt.bar(filtered_data["Date"], filtered_data["Volume"])

plt.title("Trading Volume of Alphabet Inc.")
plt.xlabel("Date")
plt.ylabel("Volume")
plt.xticks(rotation=45)
plt.grid(True)

# Display the graph
plt.show()
```

Output:



=====			
ALPHABET STOCK DATA			
	Date	Close	Volume
1	2020-01-03	67.75	1650000
2	2020-01-06	69.89	1800000
3	2020-01-07	69.76	1750000
4	2020-01-08	70.25	1900000
5	2020-01-09	71.00	2100000
6	2020-01-10	70.80	2050000
7	2020-01-13	71.50	2200000
8	2020-01-14	72.10	2150000